AERIAL ROBOTICS SYSTEM ARCHITECTURE

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Components

- Processing Unit: Microcontrollers (Arduino, STM32) or SBCs (Raspberry Pi, Jetson).
- Sensors and Actuators: Inputs and outputs for interaction with the environment.
- Communication Module: Enables data transfer and remote control.
- Power Supply: Batteries, converters, backup systems.

Communications

- Wired (USB, CAN): Reliable, low-latency for local systems.

Wireless:
- RF modules: Short-range remote control.
- Wi-Fi: Real-time data streaming and updates.
- 4G/5G: High bandwidth, supports cloud control.
- Satellite: Used in remote or marine applications.

Ground Control Station (GCS)

- Centralized Monitoring Hub: Visualizes telemetry, maps, camera feed.
- Control Input: Commands are sent to the robot via UI or code.
- Diagnostic Tools: Battery, health status, mission logs.

Thank You!